HW 4

30/10/2025

10 Points Possible

Due Oct. 30

4.1 # 7, 12, 17, 18, 19

4.2 # 1, 5

Homework 4 Problems (from 3rd edition textbook)

4.1: Matrix Algebra

7) Let A have the block form

$$A = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}$$

in which the blocks are $n \times n$. Prove that if $B - I$ is nonsingular, then, for $k \geq 1$,

$$A^k = \begin{bmatrix} B^k & (B^k - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

$$A^1 = \begin{bmatrix} \underset{n}{B} & C \\ \underset{n}{0} & \underset{n}{I} \end{bmatrix} \begin{bmatrix} \underset{n}{B} & \underset{n}{C} \\ \underset{n}{0} & \underset{n}{I} \end{bmatrix}.$$

$$A^2 = \begin{bmatrix} B^2 & BC + C \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}$$

$$A^3 = \begin{bmatrix} B^3 & B^2C + BC + C \\ O & I \end{bmatrix}$$

$$A^k = \begin{bmatrix} B^k & (B^{k-1} + B^{k-2} + \dots + I)C \\ O & I \end{bmatrix}$$

Is $(B^2 - I)(B - I)^{-1} C = BC + C$?

$S_0 = I$

$S_1 = B + I$

$S_2 = B^2 + B + I$

$S_n = B^n + B^{n-1} + \dots + B^1 + I$

$B S_{n-1} = B^n + B^{n-1} + \dots + B^1$

$(S_n - B S_{n-1}) = I$

$S_n = S_{n-1} + B^n$

$$\frac{\begin{aligned} S_n &= B S_{n-1} + I \\ B S_n &= B S_{n-1} + B^{n+1} \end{aligned}}{(B - I) S_n = (B^{n+1} - I)}$$

If $B - I$ is NS, then:

$$S_n = (B^{n+1} - I)(B - I)^{-1}$$

$$S_{k-1} = (B^{k} - I)(B - I)^{-1}$$

12) Let A be an $n \times n$ invertible matrix, and let u and v be two vectors in $\mathbb{R}^n$. Find the necessary and sufficient conditions on u and v in order that the matrix

$$\begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}$$

be invertible, and give a formula for the inverse when it exists.

$$M = \underbrace{\begin{bmatrix} A_{n\times n} & u_{n\times 1} \\ v^T_{1\times n} & 0_{1\times 1} \end{bmatrix}}_{(n+1)\times(n+1)} \qquad \text{Let } M^{-1} = \begin{bmatrix} B_{n\times n} & \hat{u}_{n\times 1} \\ \hat{v}^T_{1\times n} & k_{1\times 1} \end{bmatrix}$$

$$MM^{-1} = I = \begin{bmatrix} I_{n\times n} & 0_{n\times 1} \\ 0_{1\times n} & 1 \end{bmatrix}$$

$$\boxed{AB + u\hat{v} = I_{n\times n}} \quad (1)$$

$$\boxed{A\hat{u} + ku = 0_{n\times 1}} \quad (2)$$

$$\boxed{\hat{u} = -A^{-1}ku}$$

$$\boxed{v^T B = \vec{0}_{1\times n}} \quad (3)$$

$B = 0?$

$$\boxed{v^T u = 1} \quad (4)$$

$$\Downarrow$$

$$v^T \neq \vec{0}$$

$$-v^T A^{-1} K u = 1$$

$$\boxed{v^T A^{-1} u = -\frac{1}{K}}$$

$$u v^T = (I - AB)$$

17) A square matrix A is said to be skew-symmetric if $A^T = -A$. Prove that if A is skew-symmetric, then $x^T A x = 0$ for all x.

$A^t = -A$ $\quad \underset{2}{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\quad A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$

$a_{ij} = -a_{ji}$ $\quad -A = \begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix}$

$\left.\begin{matrix} a = -a \\ d = -d \end{matrix}\right] \to a = d = 0$

$-b = c$

$A = \begin{bmatrix} a & b \\ -b & d \end{bmatrix}$

$A^t = \begin{bmatrix} a & -b \\ b & d \end{bmatrix}$ $\quad -A^t = [$

$(x_1 \; x_2 \; x_3) \underset{3\times3}{\begin{bmatrix} 0 & \cdot & \cdot \\ \cdot & 0 & \cdot \\ a & \cdot & 0 \end{bmatrix}} \underset{3\times1}{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}} =$

$\underset{1\times3}{(x_1 \; x_2 \; x_3)} \underset{3\times1}{\begin{bmatrix} a_{12}x_2 + a_{13}x_3 \\ a_{21}x_1 + a_{23}x_3 \\ a_{31}x_1 + a_{32}x_2 \end{bmatrix}} = x_1(a_{12}x_2 + a_{13}x_3)$

$$\underbrace{\begin{bmatrix} 0 & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & 0 & & & \\ \vdots & & \ddots & & \\ a_{n1} & \cdots & & a_{n-1,n} & 0 \end{bmatrix}_{n\times n} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}$$

$$\begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} 0 + a_{12}x_2 + a_{13}x_3 + \cdots a_{1n}x_n \\ a_{21}x_1 + 0 + a_{23}x_3 + \cdots a_{2n}x_n \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{n,n-1}x_{n-1} + 0 \end{bmatrix}$$

$$a_{12}x_2x_1 + a_{13}x_3x_1 \cdots a_{1n}x_nx_1$$

$$x^T A x = k$$

$$(x^T A x)^T = k^T = k \quad \sigma \quad x^T A^T x = k$$

$$-x^T A x = k \Rightarrow k = 0 \quad \sigma$$

18) (Continuation) Prove that the diagonal elements of a skew-symmetric matrix are 0. Also, prove that the determinant is 0 when the matrix is of odd order.

19) (Continuation) let A be any square matrix, and define $A_0 = \frac{1}{2}(A + A^T)$ and $A_1 = \frac{1}{2}(A - A^T)$. Prove that A_0 is symmetric, that A_1 is skew-symmetric, that $A = A_0 + A_1$ and that for all x, $x^T A x = x^T A_0 x$. This explains why, in discussing quadratic forms, we can confine our attention to **symmetric matrices**.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & \vdots \\ \vdots & & \ddots & \vdots \\ a_{n1} & \cdots & \cdots & a_{nn} \end{bmatrix}$$

$$A_0 = \frac{1}{2}(A + A^T) = \begin{bmatrix} \ddots & \frac{a_{12}+a_{21}}{2} \\ \frac{a_{12}+a_{21}}{2} & \ddots \end{bmatrix} \quad 0$$

$$A_1 = \frac{1}{2}(A - A^T)$$

$$A = A_0 + A_1$$

$$x^T(A_0 + A_1)x = x^T A_0 x + 0$$

0

4.2: LU and Cholesky Factorizations

1) Proof these facts, needed in the proof of Theorem 2.

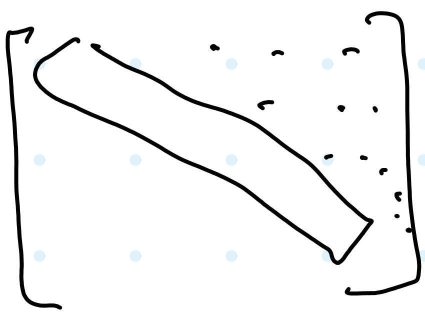
   (a) If U is upper triangular and invertible, then U^{-1} is upper triangular.

   (b) The inverse of a unit lower triangular matrix is unit lower triangular.

   (c) The product of two upper (lower) triangular matrices is upper (lower) triangular.

$$U = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix} \qquad ad \neq 0$$

$$U = \lambda I +$$



$$\begin{bmatrix} a & b \\ 0 & d \end{bmatrix} \begin{bmatrix} x & y \\ & \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

5) Prove that an upper or lower triangular matrix is nonsingular if and only if its diagonal elements are all different from 0.

$[\lambda] \neq 0$

$$\begin{bmatrix} a_{11} & 0 \\ a_{21} & a_{22} \end{bmatrix} \qquad \underbrace{a_{11}\,(a_{22} - 0)}_{\neq 0} \neq 0$$

$$\begin{bmatrix} a_{11} & & \\ a_{12} & a_{22} & \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \qquad \underbrace{a_{11}\,(a_{22}(a_{33} - 0))}_{\neq 0} \neq 0$$

$\neq 0$

$\neq 0.$